



A survey of multimodal recommender systems: methods, challenges, and future directions

Ziyu Li¹ · Shiqing Wu² · Weihua Li¹ · Quan Bai³ · Jianhua Jiang⁴

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Abstract

Multimodal Recommender Systems (MMRS) improve recommendation quality by incorporating content modalities, such as text, images, audio, and video. This survey synthesizes over 127 recent publications (2016–2025) and organizes the MMRS landscape through a novel pipeline-based taxonomy comprising six stages: modality extraction, filtration, bridging, fusion, augmentation, and optimization. Each stage is analyzed in terms of its design choices, representative methods, and practical trade-offs. Different from prior surveys that focus on model architectures or application domains, we emphasize the intermediate processing stages, i.e., filtration, bridging, and augmentation, that are critical for robustness yet under-analyzed in the literature. We further examine how generative models, large language models (LLMs), and multimodal knowledge graphs are reshaping key stages of the MMRS pipeline, and discuss open challenges including cross-domain adaptation, personalized modality preference modeling, and LLM integration. By linking design choices to evaluation practices and identifying concrete research gaps, this survey provides a structured technical roadmap for advancing MMRS.

Keywords Multimodal Recommender Systems · Survey · Deep Learning · Multimodal Fusion · Large Language Models

1 Introduction

The exponential growth of digital content has intensified information overload, making recommender systems indispensable for personalized information access [1, 2]. Tradi-

tional collaborative filtering and content-based methods rely primarily on user-item interaction histories or item metadata; despite their success, they suffer from data sparsity and the cold-start problem as platforms continually introduce new users and items with little historical data [3, 4].

To overcome these limitations, Multimodal Recommender Systems (MMRS) incorporate heterogeneous signals, e.g., text, images, audio, video, knowledge graphs, and contextual cues, into user and item representations [3, 5]. Fusing diverse modalities enables MMRS to capture richer user preferences and item semantics, yielding higher accuracy, stronger cold-start performance, and more interpretable and diverse recommendations [4–6].

The evolution of MMRS has progressed through several distinct phases. Early work extended collaborative filtering and matrix factorization with handcrafted features (e.g., BoW, SIFT, MFCC) [1], but such manual feature engineering limited the ability to capture complex multimodal patterns. The deep learning era introduced end-to-end modality-specific encoders, such as CNNs for images and RNNs and Transformers [7] for text, which significantly improved accuracy and scalability [2, 8]. Subsequently, graph-based models enabled richer relational modeling of user-item-modality interactions [9–12], while variational

✉ Shiqing Wu
sqwu@cityu.edu.mo

✉ Weihua Li
weihua.li@aut.ac.nz

Ziyu Li
ziyu.li@autuni.ac.nz

Quan Bai
quan.bai@utas.edu.au

Jianhua Jiang
jjh@jlufe.edu.cn

¹ Department of Data Science and Artificial Intelligence, Auckland University of Technology, Auckland, New Zealand

² Faculty of Data Science, City University of Macau, Macau SAR, China

³ School of Information and Communication Technology, University of Tasmania, Hobart, TAS, Australia

⁴ Jilin Province Key Laboratory of Fintech, Jilin University of Finance and Economics, Changchun, China

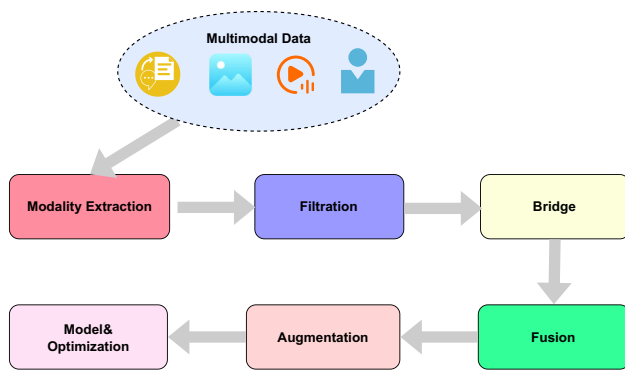


Fig. 1 Pipeline-based taxonomy for MMRS. Our six-stage framework extends prior taxonomies [3, 5] by explicitly modeling filtration, bridging, and augmentation as separate stages

inference and disentanglement techniques began to model uncertainty and cross-modal independence more explicitly [13, 14]. Subsequently, self-supervised and contrastive learning paradigms further improved representation robustness and cross-modal alignment [15–17], and generative models alongside prompt-tuning and knowledge distillation approaches enabled more data-efficient and semantically coherent multimodal fusion [18–20]. Most recently, large-scale pre-trained models, particularly multimodal LLMs, have been leveraged for sequential recommendation and deeper cross-modal alignment [21, 22].

Given the rapid evolution of the field, there is a pressing need for comprehensive surveys that synthesize recent

advances. For example, Baltrušaitis et al. [4] provide a broad multimodal ML taxonomy, Zhou et al. [5] categorize MMRS by model architecture, and Wu et al. [22] focus on LLM-based recommendation. While these prior surveys offer valuable perspectives, they rarely systematize the intermediate processing stages where many practical challenges reside: filtration (noise removal, missing-modality handling, efficiency-oriented screening), bridging (structural, interactive, mapping, and hybrid alignment), and augmentation (data-, feature-, and representation-level) [1, 23–26]. To fill these gaps, we propose a pipeline-oriented taxonomy that structures MMRS into six core stages as illustrated in Figure 1, which includes modality extraction, filtration, bridging, fusion, augmentation, and optimization (supervised, self-supervised, distillation). We link each stage to concrete design choices and evaluation practices [2, 12, 27].

Figures 2 and 3 outline the roadmap of the survey. We first detail Modality Processing, including four stages that prepare and align information before modeling: Filtration removes noise, handles missing modalities, and streamlines computation; Bridging aligns heterogeneous signals through structural, interactive, mapping, and combined strategies; Fusion integrates evidence at feature, representation, or decision level; Augmentation strengthens generalization through data-level, feature-level, and representation-level techniques. We then survey Modeling, from matrix factorization and classic deep architectures to attention, graph neural networks, and generative or LLM-based approaches, and conclude with Optimization strategies, including supervised and

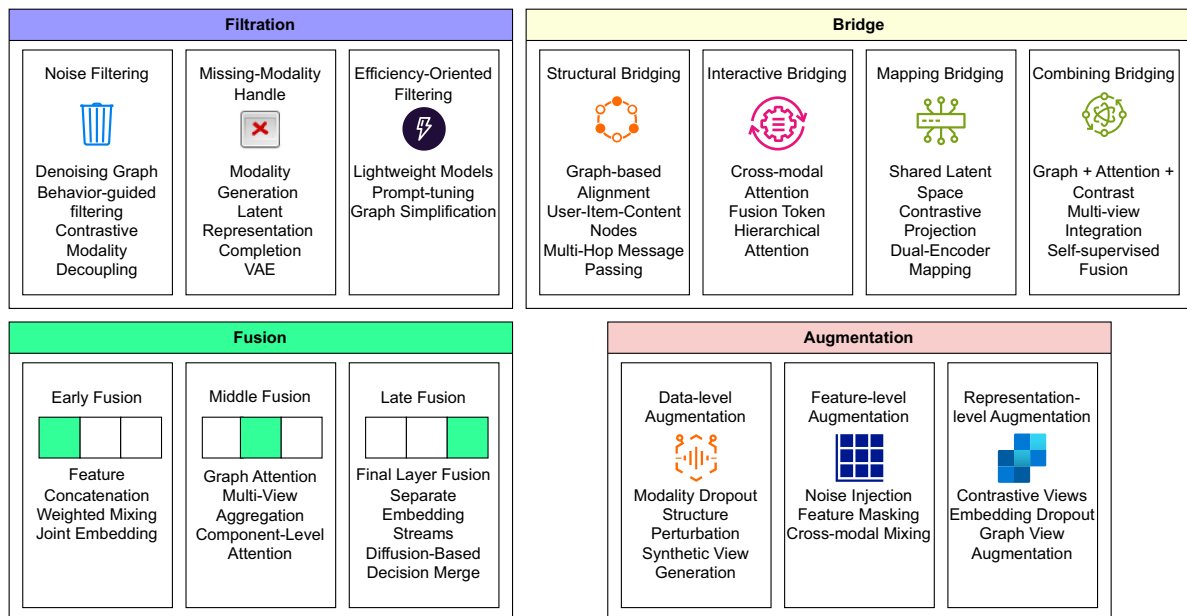


Fig. 2 MMRS modality processing roadmap, illustrating filtration, bridging, fusion, and augmentation stages. Techniques span noise filtering [15], graph-based structural bridging [9], attention-based interactive bridging [10], mapping bridging [28], and multi-level augmentation [29]

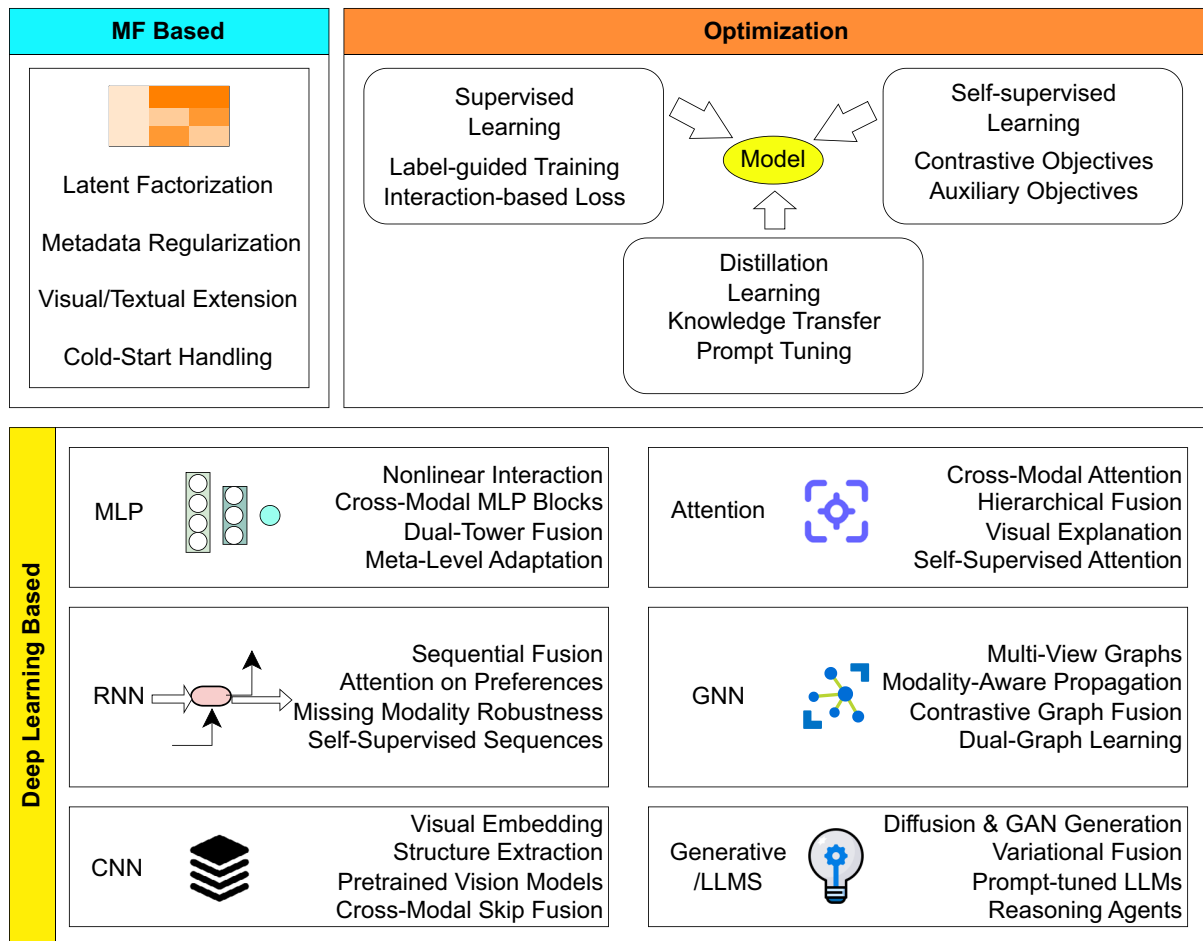


Fig. 3 MMRS model families and optimization strategies. Model architectures range from matrix factorization [8] and GNNs [9, 11] to generative and LLM-based approaches [18, 30]. Optimization covers supervised, self-supervised [29], and distillation learning [19]

self-supervised learning, as well as distillation, knowledge transfer, and prompt tuning. Together, these sections provide a coherent pipeline view that links design choices to practical trade-offs and evaluation.

The main contributions of this survey are summarized as follows:

- We introduce a new pipeline-based taxonomy for MMRS, systematically organizing the field into six key stages: modality extraction, filtration, bridging, fusion, augmentation, and optimization. This framework offers a unified perspective for analyzing diverse architectures and design choices.
- We survey the most recent literature up to 2025, capturing the latest developments and research frontiers in MMRS, including key studies presented at top-tier conferences and journals.
- We highlight the growing integration of Generative AI, particularly LLMs, into MMRS. These technologies

enhance the system's capabilities in multimodal alignment, user behavior modeling, and content generation.

- We outline future directions, including the need for more task-specific adaptation of LLMs to recommendation contexts, the development of efficient and reliable multimodal reasoning mechanisms, cross-domain generalization under modality heterogeneity, and the integration of multimodal knowledge graphs with commonsense reasoning into MMRS pipelines.

2 Preliminaries

As the foundation for our survey, this section introduces the problem formulation of MMRS and outlines the initial step of modality extraction. These preliminaries establish the conceptual and operational context for the remainder of the pipeline. We begin by formalizing the multimodal recommendation task, detailing how heterogeneous modality features contribute to modeling user-item interactions. We

then introduce common techniques for extracting informative representations from various modalities, incorporating the evolution from handcrafted features to modern deep and pre-trained encoders.

2.1 Multimodal recommendation problem definition

Given a set of users $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ and a set of items $\mathcal{I} = \{i_1, i_2, \dots, i_N\}$, the goal of a multimodal recommender system (MMRS) is to predict a user's preference r_{ui} for an item i by effectively leveraging information from multiple modalities. Each item i is associated with a set of multimodal features:

$$\mathbf{x}_i = \{\mathbf{x}_i^{(1)}, \mathbf{x}_i^{(2)}, \dots, \mathbf{x}_i^{(K)}\},$$

where $\mathbf{x}_i^{(k)}$ denotes the feature vector of the k -th modality (e.g., text, image, audio, or video) [5].

While multimodal features are predominantly associated with items, some recent work has also explored user-oriented modalities, such as profile descriptions, review histories, and behavioral traces, particularly in user modeling and conversational settings [32]. However, this remains a less common setting in MMRS, and we focus primarily on item-side multimodal features throughout this survey.

The recommendation task is typically framed as estimating a matching score between user u and item i :

$$\hat{r}_{ui} = f(\mathbf{e}_u, \mathbf{e}_i),$$

where $\mathbf{e}_u$ and $\mathbf{e}_i$ are the latent representations of the user and item, respectively. These latent representations are vector embeddings that encode semantic and behavioral characteristics derived from the available input data. $\mathbf{e}_i$ is constructed by fusing features from $\mathbf{x}_i$, and $\mathbf{e}_u$ may be learned from implicit interaction histories. In multimodal settings, the function $f(\cdot)$ captures how modality-specific components of $\mathbf{e}_i$ and $\mathbf{e}_u$ interact. For instance, dot-product functions assume a unified embedding geometry, while attention-based or MLP-based variants can model asymmetric or non-linear cross-modal dependencies. This formulation applies to both explicit feedback, such as ratings, and implicit feedback, including clicks, purchases, and dwell time. MMRS models aim to enhance performance by enriching user and item representations with multimodal content, thereby enabling more accurate, robust, and personalized recommendations.

2.2 Modality extraction

Modality extraction transforms raw multimodal content, including text, images, audio, and video, into structured feature representations that serve as inputs to downstream pipeline stages. The quality of these representations directly shapes the expressiveness of user and item embeddings. For instance, the transition from hand-crafted to CNN-based visual features produced consistent performance gains in early visual recommendation benchmarks [8], illustrating the sensitivity of downstream modeling to extraction quality.

The evolution of modality extraction in MMRS can be understood in three phases. In the earliest phase, extraction relied on hand-crafted, modality-specific descriptors: textual content was represented via Bag-of-Words or TF-IDF, visual content via SIFT keypoints, and audio signals via Mel-Frequency Cepstral Coefficients (MFCC) [1]. While interpretable and computationally lightweight, these methods failed to capture contextual or semantic relationships and generalized poorly across domains.

The introduction of deep learning marked a significant shift toward data-driven extraction. Convolutional Neural Networks (CNNs) became the standard for learning hierarchical visual features directly from image pixels [8], while Recurrent Neural Networks (RNNs) and later Transformers [7] enabled modeling of long-range dependencies in textual and sequential data [32]. These architectures supported end-to-end training and substantially improved both accuracy and scalability, though they typically required large amounts of labeled data and modality-specific architectural design choices.

More recently, large-scale pre-trained multimodal encoders, such as CLIP [28], BLIP [33], and VLMO [34], have become prominent in MMRS. Trained on massive datasets with contrastive or mixture-of-expert objectives, these models produce semantically rich, cross-modally aligned embeddings that transfer effectively to downstream recommendation tasks, including zero-shot settings [28, 33, 34]. Their application to MMRS has been explored in [35, 36], demonstrating that such encoders can substantially reduce the need for task-specific feature engineering. However, they introduce challenges related to computational cost, domain specificity, and potential modality-alignment mismatches when applied to recommendation-specific content distributions.

Table 1 summarizes the deep learning methods and pre-trained encoders, which constitute the dominant extraction paradigms in modern MMRS.

Table 1 Summary of Modality Extraction Techniques in MMRS

Technique (Year)	Contributions	Limitations
CNN-based Visual Features [31] (2015), [8] (2016)	Learn discriminative visual features for item representation; support cold-start handling via visual signals.	Static features not adapted to the recommendation context; may introduce visual bias when other modalities are more informative.
RNN/Transformer-based Text Features [7] (2017), [32] (2018)	Model long-range dependencies in text and user interaction sequences; improve content-aware and sequential recommendation.	RNNs struggle with long sequences; both require substantial domain-specific text data to perform well.
Pre-trained Multimodal Encoders (CLIP, BLIP, VLMo) [28] (2021), [33] (2022), [34] (2022)	Cross-modally aligned embeddings transferable to MMRS; reduce feature engineering overhead and support cold-start via zero-shot generalization.	Pretrained on general web data, potentially misaligned with recommendation-specific content distributions; high computational cost.

3 Modality enhancement and integration

Following the extraction of raw modality features, MMRS needs to refine, align, and integrate these representations to support effective downstream modeling. This section reviews four key operations that serve this purpose: filtration, bridging, fusion, and augmentation. Table 2 reviews filtration, bridging, fusion, and augmentation, with typical strengths and limitations and key references.

3.1 MMRS filtration

Filtration in MMRS is a critical step that ensures the quality, robustness, and scalability of the learning process. In this section, we categorize existing filtration strategies into three key types: noise filtering, missing-modality handling, and efficiency-oriented filtering.

3.1.1 Noise filtering

Noise filtering aims to suppress misleading or irrelevant content that may deteriorate the quality of multimodal recommendations. Several recent studies propose diverse noise filtering mechanisms that operate at the graph, feature, and interaction levels to mitigate such noise in multimodal data. For example, Zhou and Shen present FREEDOM, a dual-graph framework that explicitly addresses structure-level noise in MMRS by freezing the relatively stable item-item graph and denoising the user-item interaction graph, thereby restricting noisy interaction propagation and improving model robustness [15]. Yu et al. design a multi-view graph convolutional network with a behavior-guided purifier that leverages user interaction patterns to filter noisy modality signals before multimodal fusion, highlighting the importance of behavior-aware modality purification in multimodal graph learning [37]. Liu et al. introduce the EliMRec framework, which addresses single-modal bias by treating it as

noise and applying decoupling techniques to reduce inter-modal interference [38]. Yang et al. propose a modal-aware bias-constrained contrastive learning method that encourages consistency across modality representations and suppresses noisy modality signals implicitly via alignment constraints [39]. Lin et al. develop a contrastive modality-disentangled learning strategy that separates modality-specific and shared information, effectively reducing cross-modal noise and improving representation quality [40]. In addition, Tan et al. introduce an adaptive pipeline that first filters noise and then enhances modal features, further suppressing spurious signals before fusion and thereby improving robustness [41].

The existing works demonstrate that effective filtering can significantly reduce modality-specific noise and improve recommendation accuracy. However, most existing techniques are designed for specific model structures or datasets, which may limit their generalizability. Additionally, while these approaches often report improvements on benchmark datasets, they rarely provide an in-depth analysis of filtering effectiveness or adaptability across varying user contexts.

3.1.2 Missing-Modality handling

Handling missing modalities is essential for maintaining robustness when one or more input modalities are absent. This is a common scenario in real-world systems. Wang et al. propose LRMM, a model that learns the distribution of missing modalities and employs modality dropout and a multimodal sequential autoencoder to infer absent features [42]. Hu et al. propose MMSR (Multi-Modality enriched Sequential Recommendation), which features an adaptive fusion mechanism with asynchronous update gates and dual-attention graph propagation, enabling it to naturally handle missing modalities during sequential recommendation [44]. Malitesta et al. propose a graph-aware imputation pipeline, which combines classic imputers with item-item co-purchase and cross-item multimodal-similarity propagation

Table 2 Modality Enhancement and Integration Techniques in MMRS

Categories	Techniques	Contributions	Limitations
Filtration	Noise Filtering [15] (2023), [37] (2023), [38] (2022), [39] (2023), [40] (2025), [41] (2025)	Suppresses irrelevant or misleading modality input to enhance recommendation quality.	Sensitive to parameter tuning and may discard informative signals.
	Missing-Modality Handling [42] (2018), [43] (2024), [44] (2023), [45] (2024), [46] (2024)	Infers or reconstructs missing modality data to ensure model robustness.	May introduce bias or uncertainty; performance depends on the quality of generated content.
	Efficiency-Oriented Filtering [47] (2020), [48] (2023), [19] (2024), [49] (2021), [50] (2024)	Reduces computational cost while maintaining performance; useful for large-scale systems.	Risk of oversimplification or loss of model capacity.
	Structural Bridging [9] (2019), [11] (2021), [51] (2021), [52] (2022), [53] (2020), [54] (2021), [55] (2018)	Builds graphs to connect modalities and facilitate semantic propagation.	Graph construction can be expensive and domain-specific.
Bridging	Interactive Bridging [49] (2021), [10] (2020), [56] (2017), [57] (2023), [30] (2025), [58] (2019)	Uses attention and interaction modules to align modalities without unifying embeddings.	May not fully resolve semantic mismatches between modalities.
	Mapping Bridging [59] (2019), [60] (2022), [29] (2025), [20] (2024), [61] (2023)	Projects different modalities into a shared latent space using contrastive or self-supervised learning.	Requires large, aligned datasets; domain-specific fine-tuning needed.
	Combining Bridging [62] (2023), [63] (2024), [15] (2023), [18] (2024), [64] (2025)	Integrates structural, interactive, and mapping methods for more flexible fusion.	Architecturally complex; may require significant resources.
	Early Fusion [8] (2016), [13] (2021), [65] (2017), [66] (2016), [67] (2021)	Combines raw or low-level features for direct multi-modal interaction.	Shallow integration may limit modality specialization.
Fusion	Middle Fusion [56] (2017), [58] (2019), [10] (2020), [51] (2021), [49] (2021), [37] (2023), [68] (2024), [69] (2024), [70] (2025)	Fuses modality-specific embeddings via attention or alignment at intermediate layers.	Complex to tune; may struggle with noisy modalities.
	Late Fusion [42] (2018), [71] (2025), [41] (2025), [30] (2025)	Keeps modalities independent and combines outputs at the final prediction stage.	Limited interaction across modalities; late decisions may miss deep synergy.
	Data-level Augmentation [42] (2018), [72] (2022), [73] (2023), [64] (2025)	Alters raw inputs or graph structure to simulate missing data and improve robustness.	Might introduce unrealistic scenarios or distortions.
	Feature-level Augmentation [74] (2022), [29] (2025), [75] (2017), [13] (2021), [76] (2025), [77] (2023)	Applies noise, masking, or transformation to intermediate modality features.	Excessive perturbation may degrade feature integrity.
Augmentation	Representation-level Augmentation [78] (2023), [29] (2025)	Generates diverse latent representations using contrastive and multi-view learning.	Alignment of augmented views can be unstable if the data is sparse.

to reconstruct absent features [45]. Ganhör et al. introduce a single-branch, weight-sharing embedding network that encodes different modalities with a single encoder, remains functional when some modalities are absent, and yields strong results in cold-start and missing-modality scenarios [46]. Yi and Chen propose a variational mixture-of-experts model that selectively activates modality-specific experts, naturally accommodating absent modalities by relying on the remaining active experts [43].

These recent advances in handling missing modalities demonstrate notable progress by leveraging adaptive fusion, graph-based imputation, and modality-agnostic encoding techniques. These methods effectively infer missing modality features by learning cross-modal dependencies and latent structures. However, a key limitation is that many approaches rely on strong assumptions about the distributional alignment between modalities, which may not hold in highly heterogeneous or sparse real-world datasets. Furthermore, imputation and reconstruction methods may introduce bias if the learned distributions do not faithfully reflect true modality content, and weight-sharing encoders may sacrifice modality-specific expressiveness for robustness.

3.1.3 Efficiency-Oriented filtering

Efficiency-oriented filtering aims to reduce computational and memory costs while preserving or improving recommendation accuracy, a crucial consideration in large-scale MMRS deployments. In this context, filtering is not limited to noise suppression; it also involves retaining only the most informative structures, parameters, or cross-modal interactions to avoid unnecessary computation. Existing studies and related efficiency-oriented techniques have explored this objective from multiple perspectives, including structural simplification, engineering streamlining, model compression, and selective information routing. For example, although not specifically designed for MMRS, LightGCN simplifies graph convolutional recommendation by removing feature transformations and nonlinear activations, thereby reducing computational complexity while retaining the essential neighborhood aggregation mechanism [47]. Wei et al. propose PromptMM, a prompt-tuning and distillation-based framework that compresses large models into lightweight alternatives with limited performance loss, offering an efficient solution for multimodal recommendation under resource constraints [19]. Similarly, attention bottlenecking has been explored as a general multimodal efficiency technique. Nagrani et al. introduce an attention bottleneck mechanism that restricts information exchange to a compact set of essential attention pathways, thereby reducing the cost of multimodal fusion [49]. More directly within MMRS, Guo et al. develop LGMRec, which balances local and global graph learning by selectively emphasizing informative graph

structures, improving recommendation effectiveness without incurring excessive structural complexity [50].

Existing methods indicate that efficiency-oriented filtering in MMRS can be achieved through different but complementary strategies, ranging from simplifying model architectures and restricting information flow to compressing large models and retaining only essential graph structures. These mechanisms can substantially reduce model size, computation, or inference cost while maintaining competitive recommendation performance. However, such methods still involve trade-offs between efficiency and modeling capacity. For example, structural simplification, as in LightGCN, may weaken the ability to capture complex feature interactions, while compression-based methods such as PromptMM often depend on large pretrained models, which may themselves be costly and potentially introduce fairness or bias concerns if not carefully controlled. Moreover, most existing efficiency-oriented methods are validated on fixed benchmark datasets, with limited evidence regarding their scalability across heterogeneous domains or their adaptability to evolving user preferences and modality conditions.

3.2 Modality bridging

In MMRS, modality bridging plays a crucial role in addressing the inherent heterogeneity across data types such as text, images, audio, and video. Its primary goal is to reduce the semantic gaps between modalities, enabling effective alignment and collaboration of information before final prediction [79]. Bridging mechanisms ensure that multimodal signals interact meaningfully, thereby facilitating richer, more coherent user-item representations.

3.2.1 Mapping bridging

Mapping bridging is one of the most commonly adopted strategies in MMRS. It aligns modality-specific features by projecting them into a shared latent space, thereby enabling inter-modal comparisons and joint representation learning. In multimodal recommendation, this strategy is widely used to bridge heterogeneous item content and collaborative signals, often through contrastive learning, embedding alignment, or self-supervised objectives to encourage semantic consistency across modalities. More broadly, similar alignment paradigms have also been extensively studied in general multimodal learning, such as contrastive image-text embedding and unified cross-modal representation learning [28, 80, 81].

In the recommendation context, VBPR maps visual features into the collaborative filtering space to address cold-start issues [8]. Han et al. separate modality pretraining from alignment to improve flexibility [60]. Extending this idea to multimodal content, MAML integrates textual and visual features into a unified item representation and further

applies attention to capture item-specific user preferences. Despite the use of attention, its core bridging mechanism remains mapping-oriented, since multimodal signals are first fused into a shared representation space [59]. MENTOR applies self-supervised learning to map heterogeneous features into a shared representation space [29]. Align-Rec decomposes training into three alignments, i.e., within content, content-ID, and user-item, using a pretrain-then-joint-train schedule to improve cross-modal consistency [20]. Wei et al. employ multi-level intra- and inter-modal contrastive learning to reinforce shared semantic structure [61]. BM3 follows a self-supervised approach by bootstrapping latent views, reconstructing the user-item graph, and enforcing inter-/intra-modal contrastive alignment, without relying on auxiliary graphs, thereby learning a unified representation space for users and items [78].

Despite its wide applicability, mapping bridging has inherent limitations. Projecting all modalities into a single latent space assumes a semantically coherent embedding geometry, which may not hold when modalities are highly divergent in structure or quality. Moreover, aggressive alignment can risk modality collapse, where the unique characteristics of each modality are lost, degrading downstream performance. Furthermore, mapping-based methods often underperform in scenarios with asynchronous or missing modalities, as the shared space may fail to compensate for the absence of modality-specific cues.

3.2.2 Structural bridging

Structural bridging constructs inter-modality relationships by modeling multimodal data through graph-based structures. These methods represent users, items, and modality-specific content (e.g., images, text) as nodes, and encode behavioral or semantic associations via edges.

Several models employ this approach. For example, MMGCN constructs separate graphs for each modality and connects them via user-item interactions to support cross-modal learning [9]. DualGNN enhances relational context by combining user-item and user-user graphs in a dual-graph framework [11]. LGMRec adopts a hybrid approach that incorporates both local and global graphs to capture fine-grained and holistic patterns [50].

Some approaches emphasize item-item relationships. LATTICE learns modality-aware item-item graphs from multimodal content and then aggregates them [51], while HCGCN performs dynamic graph clustering over user-item and item-item graphs to model preferences more efficiently [52]. Knowledge graph-based models further expand structural bridging. MKGAT integrates multimodal content into a structured knowledge graph and uses attention mechanisms for information propagation [53]. PMGT combines item-item relations with modality features via a pre-trained

graph encoder [54], and GraphCAR employs a unified graph autoencoder to jointly model user-item interactions and content [55].

Despite their success, structural bridging approaches face several limitations. Graph construction often depends on pre-defined similarity functions, interaction frequencies, or clustering heuristics, which may not generalize well across datasets or modalities. Moreover, maintaining multiple modality-specific graphs increases computational complexity and memory usage, posing scalability challenges for large-scale systems. Additionally, static graph structures may struggle to capture temporal dynamics or adapt to evolving user preferences and content trends.

3.2.3 Interactive bridging

Interactive bridging enables cross-modal collaboration by introducing attention mechanisms or interaction modules that dynamically allow modalities to influence one another while preserving their distinct representational spaces. Unlike mapping-based strategies that project modalities into a shared latent space, interactive bridging maintains modality-specific embeddings, facilitating selective and task-dependent information exchange.

Several models illustrate this approach. For example, MGAT [10] introduces a graph attention network that adaptively learns modality-aware weights for user and item neighbors, allowing different modalities to contribute with varying importance. ACF [56] applies hierarchical attention over both item components (e.g., words in a review or image regions) and the entire item, refining user preferences based on granular and holistic signals.

Other models utilize iterative attention or fusion mechanisms. TMFUN refines inter-modal alignment via attention-guided multi-step fusion, deepening cross-modal interaction [57]. MMALFM [58] performs modality-specific attention followed by a learned fusion strategy, ensuring each modality's contribution is appropriately weighted during recommendation. MMAgentRec [30] applies cross-attention to align textual and visual modalities before integrating them within an agent-based decision framework.

Although interactive bridging provides greater flexibility and robustness, it also presents several challenges. Attention-based designs can be computationally expensive, especially when applied in multi-step or hierarchical forms. Furthermore, these methods often require careful tuning to prevent one modality from dominating others during fusion. They may also suffer when modalities are highly imbalanced in quality or availability, as the attention mechanisms may overfit to dominant signals or noise.

3.2.4 Combined bridging

Combined bridging integrates multiple modality-bridging strategies, including structural, interactive, and mapping-based, to achieve more expressive, robust, and adaptable multimodal representation learning. This hybrid paradigm leverages the complementary strengths of each bridging approach to achieve both semantic alignment and structural fidelity in complex multimodal settings.

A representative example is 3MVGAT, which combines multi-view knowledge graph attention over modality and knowledge views to support multimodal reasoning [62]. Similarly, TMF leverages the representational power of LLMs to fuse user interactions, item features, and auxiliary content [63].

Other hybrid models apply more task-specific combinations. FREEDOM freezes the item-item graph and denoises the user-item graph via degree-sensitive edge pruning, blending static structural constraints with dynamic representation refinement [15]. DiffMM [18] and DiffCL [64] extend this idea with diffusion-based learning. Specifically, DiffMM integrates modality-aware graph diffusion with cross-modal contrastive training, whereas DiffCL uses diffusion to generate contrastive views and introduces an item-item graph to mitigate sparsity. For explicit missing-modality handling within MMRS, recent work treats absent multimodal features as missing node features and propagates them on an item-item graph as a pre-processing step [45]. MONET can be categorised as a combined bridging method, as it jointly integrates multimodal representation learning (mapping) with graph-based interaction modeling (structural), thereby leveraging both representation-level and structural bridging mechanisms [69].

Despite their advantages, combined bridging approaches still present limitations. They often introduce increased model complexity, making them harder to train, tune, and interpret. Integration across bridging paradigms can also lead to optimization conflicts. For example, aligning modalities in a shared space while simultaneously preserving graph structure may require careful balancing. Moreover, scalability becomes a concern when multiple modules are integrated into a single framework, especially in large-scale recommendation settings.

3.3 Fusion strategies

Fusion refers to the process of integrating features from different modalities to construct unified representations of users and items in MMRS. Selecting an appropriate fusion strategy has a significant impact on model performance, scalability, and interpretability. Based on the stage at which modality-specific features are combined, fusion strategies

are commonly categorized into early, middle, and late fusion [82].

3.3.1 Early fusion

Early fusion combines raw or low-level features from multiple modalities at the input stage, typically via concatenation or shallow transformation [4, 82]. This approach enables early interaction between modalities and is computationally efficient. However, it may be sensitive to noisy or imbalanced modalities.

VBPR integrates image-based visual features into the matrix factorization framework, augmenting item embeddings with CNN-derived visual signals before interaction modeling [8]. MMVGAE uses modality-specific variational encoders to learn a shared latent Gaussian variable from visual and textual inputs, effectively fusing modalities via variational inference before graph decoding [13]. VISTA combines CNN-based visual, social, and temporal features at the encoder input before modeling user-item interactions, representing a typical feature-level early-fusion approach [66]. DeepStyle merges CNN-extracted aesthetic and style embeddings with collaborative filtering signals at the encoder input to capture visually-informed user preferences [65]. IMRec combines CNN-extracted visual impression features from news article screenshots with textual title embeddings at the input stage, employing local and global impression modeling to exploit the visual layout of news content for early multimodal interaction [67].

3.3.2 Middle fusion

Middle fusion involves separate processing of each modality through dedicated encoders, with fusion occurring at intermediate layers [4]. This allows modality-specific encoding before interaction, often leading to more expressive and context-aware fusion.

MGAT constructs modality-aware graph attention networks where each modality is first encoded separately, followed by joint modeling via attention-based fusion [10]. Multi-View GCN processes user-item interactions across different modalities using modality-specific graphs and integrates them via behavior-guided alignment and fusion [37]. MMALFM adopts a middle-fusion strategy by first learning modality-specific representations from review text and images, and then integrating them into a shared latent factor space to construct interpretable item representations before interaction modeling [58]. LATTICE learns modality-specific item graphs and integrates collaborative signals in a middle fusion stage to enrich multimodal representations [51]. SMORE aligns modality-specific representations in the frequency domain before fusing them at an intermediate level, aiming to improve the semantic coherence of cross-

modal fusion [70]. MONET applies dual attention modules, content-based and user-specific attention, to refine modality features before middle-stage integration [69]. Attention bottleneck mechanisms, as used in video-language models [49], restrict information exchange between modalities to predefined attention channels, enabling controlled and interpretable middle fusion. Earlier works, such as ACF [56], use component-level attention to weight modality contributions during intermediate fusion stages.

3.3.3 Late fusion

Late fusion maintains modality processing streams largely independent, combining only the final outputs for prediction [4, 82]. This method enables modular design, allowing modality-specific models to operate with minimal interference, making it particularly useful for scenarios involving missing modalities or heterogeneous data types.

LRMM processes each modality through a separate sequential autoencoder and combines the resulting modality-specific representations at the output stage, employing modality dropout during training to maintain robustness when modalities are absent [42]. AMMRM first filters noise and enhances each modality, constructs modality-level item and user-item graphs, and then applies a user-preference adaptive fusion gate to merge modal contributions before scoring, i.e., fusion happens near the output but after modality-specific refinement rather than as a single final sum [41]. MARGO follows a late-fusion scheme with modality-specific predictors and learns modality weights supervised by a modality-reliability vector; the weighted combination at inference produces the final score [71]. MMAgentRec processes text/visual/behavioral inputs with multimodal modules and then employs an LLM to orchestrate recommendation decisions at inference, effectively performing late-stage integration of the modalities within an agentic decision loop [30].

3.3.4 Comparison

Fusion strategies in MMRS determine how effectively multimodal signals are integrated, with early, middle, and late fusion each offering distinct trade-offs [4, 82]. Early fusion is computationally efficient, enabling immediate cross-modal interaction. However, it can be vulnerable to noise and modality imbalance due to its limited pre-processing capabilities. Middle fusion provides richer representational power by encoding each modality independently before integration, often using attention or alignment mechanisms. However, it increases architectural complexity and may require careful tuning to avoid overfitting or modality dominance. Late fusion preserves modality-specific information via modular processing pipelines and is particularly advantageous when

modalities are missing or optional. Yet, it often misses opportunities for deeper cross-modal synergy.

Many existing fusion methods heavily rely on benchmark datasets with fixed modality availability, which limits their generalizability to real-world settings where modality presence and quality are dynamic [5]. Moreover, few models explicitly address modality uncertainty or provide interpretability in how different modalities contribute to the final prediction. Emerging trends such as diffusion-based augmentation for fusion [18] and prompt-tuning-based knowledge distillation [19] show promise in improving fusion quality and efficiency, respectively, but at the cost of increased design complexity.

3.4 Augmentation

Augmentation techniques can help to improve the generalization, robustness, and adaptability of MMRS, especially in real-world scenarios characterized by data sparsity, noisy inputs, and missing modalities [23]. These methods are typically categorized into three levels: data-level, feature-level, and representation-level augmentation.

3.4.1 Data-level augmentation

Data-level augmentation operates directly on the raw input or graph structure, simulating imperfect conditions that often occur in practice. Techniques at this level include modality dropout, partial input masking, and graph perturbation, which are designed to expose models to diverse and incomplete inputs. For instance, LRMM introduces modality dropout by randomly omitting entire modalities during training, helping the model generalize under partial observability [42]. MMGCL incorporates modality edge dropout and modality masking to simulate noisy and sparse user-item graphs [72].

Generative methods have advanced data-level augmentation by synthesizing realistic input variants. DiffCL [64] leverages diffusion models to generate contrastive views and implements semantic alignment, which can be regarded as generative data-level augmentation. Lin et al. employ contrastive intra- and inter-modality generation to construct cross-modal views for incomplete multimedia recommendation [73].

While effective, data-level augmentation must be carefully controlled since aggressive perturbations may introduce unrealistic patterns that harm downstream performance or hinder convergence.

3.4.2 Feature-level augmentation

Feature-level augmentation introduces variation at the level of modality-specific feature vectors, offering more targeted regularization than data-level methods. Common strategies

include feature masking, corruption, dropout, and noise injection.

Tao et al. [74] adopt a self-supervised framework in which feature-noise perturbation creates augmented views, helping the model handle incomplete or noisy inputs. MENTOR uses multi-level self-supervised objectives to enhance and align features across modalities, strengthening representations from both the graph and feature perspectives [29].

Generative and reconstruction methods further enhance feature quality. For instance, Kang et al. use generative image models to synthesize plausible visual content, learning fashion-aware representations applicable to both visual recommendation and item design generation [75]. MMV-GAE applies variational inference over a multimodal graph to learn latent variables and reconstruct modality features during training [13]. Recent work also explores diffusion and contrastive objectives at the feature layer: Li et al. propose a modality-diffused counterfactual framework that completes or generates features under incomplete modalities [76], and Liu et al. construct multiple graph views for contrastive learning to strengthen cross-modal alignment and robustness [77].

Compared to data-level augmentation, feature-level methods provide finer control and often better preserve modality semantics, though they still require careful calibration to avoid over-smoothing or introducing uninformative noise.

3.4.3 Representation-level augmentation

Representation-level augmentation modifies latent embeddings directly to promote semantic consistency, alignment, and robustness within the shared representation space. These methods are often integrated with contrastive learning or multi-view objectives.

A representative example is BM3 [78], which performs bootstrap contrastive learning by selectively masking parts of the latent representation, thereby simulating semantic variability while preserving the overall embedding structure. Similarly, MENTOR employs multi-level self-supervised objectives to progressively refine embeddings from fine-grained feature representations to coarse-grained graph representations, improving robustness under modality heterogeneity [29].

While powerful, representation-level augmentation depends heavily on well-designed contrastive objectives and balancing intra- and inter-modal variability. Poorly structured perturbations at this level can lead to representational collapse or redundant views.

4 MMRS models

MMRS employs a wide range of modeling paradigms to capture complex interactions between users, items, and diverse

modality signals [2, 5]. This section reviews representative architectures spanning matrix factorization, deep neural networks, graph-based learning, generative frameworks, and LLM-enhanced approaches. Table 3 overviews major model families and their core advantages and limitations.

4.1 Matrix factorization-based models

Matrix factorization (MF) provides a foundational framework in recommender systems by decomposing the user-item interaction matrix into latent factors [83]. Numerous extensions have been proposed to integrate side information and address the challenges of cold starts. For example, Agarwal and Chen introduce a regression-based MF model with metadata regularization [84], while Rendle proposes Factorization Machines to model arbitrary pairwise feature interactions [85].

Early work on multimodal integration coupled MF with topic models. Wang and Blei jointly model ratings and textual content via MF and Latent Dirichlet Allocation [86], later extended by McAuley and Leskovec, who align rating dimensions with textual topics [87]. Visual signals have also been incorporated into MF-style models. VBPR extends BPR-MF with learnable visual factors for visually informed ranking [8], and CDL jointly trains a stacked denoising autoencoder with probabilistic MF, coupling content representation learning and collaborative filtering in a unified optimization framework [89].

Despite their practicality, MF-style models mostly assume linear interactions within a shared latent space, which underfit complex cross-modal and temporal dependencies and can induce modality collapse when heterogeneous signals are forced into a single set of factors. Even with Factorization Machines, the emphasis on pairwise terms leaves higher-order structure and dynamics largely unmodeled, and performance is sensitive to sparsity, missing modalities, and noisy side information. Coupling MF with high-dimensional encoders also raises scaling challenges and risks of dominant-modality bias, where one modality disproportionately governs the shared latent factors. These limitations motivate a shift to deep learning-based models that learn non-linear interactions using modality-specific encoders, attention mechanisms, and sequential architectures.

4.2 Deep learning-based models

Deep learning architectures form the backbone of modern MMRS, offering modality-specific encoding capabilities that overcome the linear assumptions of MF-based approaches. Each architectural family addresses different aspects of the MMRS pipeline: MLPs model nonlinear cross-modal interactions, RNNs capture sequential preference dynamics, CNNs extract visual semantics, attention mechanisms enable

Table 3 Summary of Multimodal Recommendation Model Types

Model Type	Contributions	Limitations
Matrix Factorization Models [83] (2009), [84] (2009), [85] (2010), [86] (2011), [87] (2013), [88] (2014), [89] (2015), [8] (2016)	Incorporate side information, address cold-start via topic modeling, trust networks, and visual content.	Limited expressiveness; rely on linear assumptions and sparse interaction matrices.
MLP-based Models [90] (2023), [91] (2023), [92] (2025)	Enable nonlinear interaction modeling; allow personalization through deep architectures.	May struggle with modality alignment; require significant parameter tuning.
RNN-based Models [32] (2018), [93] (2022)	Model temporal and sequential dynamics in multimodal signals.	Suffer from gradient issues and limited scalability.
CNN-based Models [8] (2016), [31] (2015), [65] (2017), [66] (2016), [67] (2021)	Extract visual patterns; capture spatial correlations in image-based content.	Limited temporal modeling; visual bias if modality-agnostic.
Attention-based Models [56] (2017), [94] (2019), [95] (2020), [96] (2022), [57] (2023)	Enhance fine-grained modality fusion; support explainability and adaptive learning.	Sensitive to modality noise; high computation cost.
GNN-based Models [9] (2019), [11] (2021), [10] (2020), [54] (2021), [19] (2024), [97] (2024), [98] (2024), [61] (2023), [69] (2024), [72] (2022), [99] (2025)	Capture structural, relational, and multi-hop interactions across modalities.	May over-smooth or overfit in sparse data; graph construction is complex.
Generative Models [75] (2017), [13] (2021), [100] (2024), [43] (2024), [64] (2025), [18] (2024), [101] (2024), [73] (2023)	Address sparsity and long-tail effects; generate missing or synthetic modality content.	Hard to train; may suffer from mode collapse or semantic drift.
LLM-based Models [63] (2024), [30] (2025), [102] (2025), [22] (2024)	Leverage pretrained language models for semantic reasoning and cross-modal understanding.	Require large-scale data and computing resources; interpretability and adaptation are ongoing challenges.
Other Approaches [28] (2021), [40] (2025), [103] (2025), [78] (2023)	Explore contrastive and hybrid paradigms; enable zero-shot and agent-based recommendation.	May lack coherence or generality; evaluation is still emerging.

adaptive modality weighting, and GNNs propagate relational signals across user-item-modality graphs.

4.2.1 MLP-based models

Multi-Layer Perceptrons (MLPs) have emerged as an increasingly important modeling paradigm in MMRS, owing to their flexibility for nonlinear transformations, multimodal feature interactions, and representation fusion. Early MLP-based studies primarily treated MLPs as multimodal fusion backbones. For instance, Liang et al. proposed a pure-MLP architecture with three dedicated components, namely the Feature Mixer Layer, the Fusion Mixer Layer, and the Prediction Layer, to capture and integrate multimodal item features for sequential recommendation [90]. Building upon this idea, MetaMMF further advances MLP-based multimodal fusion by introducing meta-learning strategies that dynamically adapt the fusion function for each item, thereby enabling more personalized modality integration [91]. More recently,

Huang et al. extended the role of MLPs beyond feature fusion and showed that they can also serve as topology-aware relational encoders in multimodal recommendation, suggesting that MLP-based frameworks are evolving into scalable and expressive architectures for modeling not only cross-modal interactions but also higher-order structural dependencies in MMRS [92].

4.2.2 RNN-based models

Recurrent neural networks (RNNs), including GRUs and LSTMs, provide a natural backbone for MMRS when modeling users' sequential behaviors and temporally evolving preferences. Early work mainly employed recurrent architectures to integrate multimodal item content into sequential recommendation. For example, MV-RNN models user preferences through GRUs operating over latent, visual, and textual channels simultaneously, enabling multi-view sequential modeling that captures complementary aspects of user pref-

erence dynamics [32]. This recurrent paradigm has also been extended to domain-specific recommendation settings. In the education domain, Ren et al. propose an LSTM- and attention-based framework to extract and fuse multimodal course information, showing the effectiveness of recurrent multimodal modeling for course recommendation [93].

4.2.3 CNN-based models

CNN-based MMRS can be broadly categorized according to how convolutional architectures contribute to multimodal recommendation. A first category treats CNNs as visual feature extractors that provide auxiliary image representations for recommendation. For example, VBPR leverages pretrained CNN features from product images and incorporates them into matrix factorization, thereby improving item ranking and alleviating cold-start issues [8]. A second category uses CNN-derived visual representations to model item relations and style semantics. McAuley et al. employ image-based representations to uncover substitute and complementary relationships among products, showing that CNNs can support relation-aware recommendation beyond simple feature enhancement [31]. A third category integrates CNN-extracted visual information with preference modeling in specific recommendation scenarios. DeepStyle learns style features of items and user preferences for visual recommendation [65], while VISTA incorporates visual and temporal signals to characterize user behavior in online art communities [66]. Finally, CNN-based multimodal recommendation has also been extended to broader content-driven domains. In news recommendation, CNN-modeled visual impressions are used together with textual signals to enrich multimodal content understanding and recommendation [67].

4.2.4 Attention-based models

Attention mechanisms have become a central design principle in MMRS, as they offer an effective way to filter noisy multimodal inputs and dynamically regulate the contribution of different modalities to recommendations. Early studies mainly employed attention for feature selection and fusion. For example, ACF introduces item-level and component-level attention mechanisms to model user preferences over multimedia content [56]. Extending this idea to more heterogeneous recommendation settings, AMNN leverages attention to jointly model textual and visual information for hashtag recommendation, illustrating the effectiveness of attention-based multimodal representation learning in social media scenarios [95]. Building on such cross-modal modeling, Zhou et al. develop a multi-step attention network that progressively integrates and refines modality-specific representations, thereby improving the exploitation of complementary information across modalities

[57]. Beyond representation learning, attention has also been used to enhance personalization and interpretability. In this vein, personalized attention mechanisms have been incorporated into visually aware recommendation models to align salient visual regions with user preferences and generate more explainable recommendations [94]. More recent studies further extend attention from fusion and explanation to fine-grained multimodal interaction modeling. For instance, DM²S² employs modality-aware attention layers to capture nuanced dependency patterns both within and across modalities, highlighting the broader role of attention as a unified mechanism for multimodal filtering, alignment, and interaction learning in MMRS [96].

4.2.5 GNN-based models

GNN-based MMRS can be broadly characterized from three perspectives: how multimodal recommendation graphs are constructed, how modality-specific information is propagated and fused over graph neighborhoods, and how graph representations are further enhanced for semantic robustness and cross-modal consistency. A first line of work focuses on graph construction for multimodal recommendation. For example, MMGCN builds modality-specific user-item graphs and performs joint graph convolution to capture high-order collaborative dependencies under multimodal settings [9]. DualGNN further extends this paradigm by introducing an additional user-user preference similarity graph alongside the user-item interaction graph, thereby enriching structure-aware multimodal representation learning through multi-relational graph modeling [11].

A second line of work emphasizes modality-aware propagation and fusion, based on the observation that different modalities and graph neighbors contribute unequally during message passing. MGAT addresses this issue by incorporating modality-aware attention into neighborhood aggregation [10], while MONET further combines target-aware attention with modality-aware graph convolution to enable more adaptive multimodal fusion during recommendation [69].

A third line of research moves beyond graph propagation itself and seeks to enhance graph representations through pre-training, self-supervision, and noise-robust multimodal learning. PMGT is a representative example, employing a graph transformer over an item-item graph and introducing pre-training objectives such as graph reconstruction and masked multimodal content modeling to bridge structural dependencies and global semantics [54]. Similarly, MMSSL and MMGCL improve representation quality through graph-based self-supervised and contrastive learning, strengthening cross-modal alignment and robustness under noisy multimodal environments [61, 72].

Recent studies have further addressed the challenges of noisy fusion and modality imbalance, and model efficiency in

multimodal recommendation. PromptMM further improves efficiency by distilling multimodal knowledge from a graph-based teacher into a lightweight LightGCN student [19]. SynerGraph incorporates textual information and attention-based fusion to suppress cross-modal noise during graph learning [97], whereas MDE explicitly balances cross-modal consistency and modality-specific distinctiveness through a node-level enhancement mechanism [99]. Meanwhile, UGT reflects a broader trend toward unified architectures by integrating multi-path Transformers with GNN backbones for more structured multimodal relational modeling [98]. Overall, GNN-based MMRS have evolved from graph-construction and multimodal-propagation frameworks into more robust, semantically enriched architectures for modeling complex structural and cross-modal dependencies.

4.3 Generative and LLM-based models

Early approaches often focused on generating visual content. For instance, Kang et al. proposed a conditional GAN framework that combines visual content with user preference signals to generate user-specific product images, thereby mitigating the cold-start problem in fashion recommendation [75]. MMVGAE combines visual and textual information through variational inference on a multimodal graph, producing a shared latent variable for each node [13]. More recent work explores disentanglement, in which Zhou and Miao develop a dual-path variational autoencoder that separates content and behavioral factors for improved interpretability [100]. Yi and Chen introduce a variational mixture-of-experts that captures multimodal distributions more effectively by selectively activating modality-specific experts [43].

Diffusion-based models use iterative denoising to generate semantically coherent samples or views. DiffCL applies diffusion processes to construct augmented views across visual and textual modalities for contrastive learning, thereby enhancing cross-modal semantic alignment and improving representation quality [64]. DiffMM combines diffusion-based augmentation with cross-modal contrastive objectives to better align multimodal content with collaborative filtering signals [18]. Ma et al. propose a multimodal conditioned diffusion model that uses observed features to complete or align missing content [101]. Lin et al. build intra- and inter-modality generative views for contrastive learning, strengthening robustness when modalities are absent [73]. CCDRec uses curriculum-conditioned diffusion to model distribution-level correlations and stabilizes training through reverse-process sampling [105]. MuSiC learns target-domain feature distributions from multimodal signals and generates target-domain user features via diffusion, handling cross-domain cold-start scenarios [106].

LLMs have also emerged as effective tools for multimodal fusion and reasoning. TMF integrates user behav-

ior sequences, item content, and auxiliary graph-structured modality features as a unified input to an LLM, enabling joint cross-modal reasoning over heterogeneous recommendation signals [63]. MMAgentRec combines images, text, and user dialogue with attention layers, residual modules, and graph-based reasoning to handle multi-turn interactions within an LLM-guided framework [30]. Jiang et al. formulate LLMs as universal retrievers for recommender systems and introduce a unified generative retrieval framework that transforms diverse recommendation tasks into a standardized input–output format. The framework also incorporates multimodal item representations, highlighting its effectiveness and scalability in industrial-scale settings [102]. More recently, Huang et al. extend this direction toward fully agentic systems, where multimodal LLMs act as interactive decision-makers capable of multi-turn recommendation reasoning [103]. These models highlight the growing convergence among generative modeling, contrastive learning, and language-model-based reasoning in MMRS.

Table 4 maps representative MMRS papers to their bridging and fusion strategies. The overview reveals that the overwhelming majority of surveyed methods adopt middle fusion, while early and late fusion remain comparatively rare, suggesting that the early/middle/late distinction may benefit from further refinement in future taxonomies.

5 Optimization

Optimization strategies in MMRS have evolved to address the unique challenges posed by heterogeneous modalities, incomplete data, and imbalanced supervision. Broadly, three dominant paradigms have emerged: supervised learning, self-supervised learning, and distillation-based learning. Each contributes complementary strengths in improving model robustness, generalization, and data efficiency. Table 5 contrasts supervised learning, self-supervised learning, and distillation-based learning in MMRS.

5.1 Supervised learning

Supervised learning remains the most established paradigm in multimodal recommendation, relying on observed user-item interactions (e.g., clicks, ratings, purchases) to guide model training. These approaches are particularly effective in data-rich environments, where sufficient interaction signals exist to supervise complex multimodal fusion.

For example, VBPR [8] extends Bayesian Personalized Ranking by integrating image features into the preference modeling process, enabling visually-informed ranking in cold-start item scenarios. DualGNN [11] constructs dual relational graphs, capturing user-user preference similarity, and leverages supervised signals to propagate and refine multi-

Table 4 Overview of Representative MMRS by Bridging and Fusion Strategy

Model	Year	Bridging	Fusion	Key Contribution
VBPR [8]	2016	Mapping	Early	Visual-aware ranking by integrating CNN image features into MF/BPR
VISTA [66]	2016	Mapping	Early	Fuses visual, social, temporal signals at encoder input for art recommendation
ACF [56]	2017	Interactive	Middle	Early “attention + multimedia” CF with item- and component-level attention
LRMM [42]	2018	Combined	Late	Modality dropout + multimodal sequential autoencoder for feature imputation (missing modalities)
MMGCN [9]	2019	Structural	Middle	Modality-specific user-item graphs with joint graph convolution
MGAT [10]	2020	Interactive	Middle	Modality-aware graph attention over neighborhoods
DualGNN [11]	2021	Structural	Middle	Dual graphs: user-item and user-user preference similarity
LATTICE [51]	2021	Structural	Middle	Content-learned modality-aware item-item graphs integrated with CF
MMGCL [72]	2022	Structural	Middle	Modality-masked / edge-drop augmentations for graph contrastive learning
BM3 [78]	2023	Mapping	Middle	Bootstrap contrastive with latent view dropout; graph reconstruction + inter/intra-modal alignment
FREEDOM [15]	2023	Combined	Middle	Freeze content-derived item-item graph; denoise user-item via degree-sensitive pruning
MGCL [77]	2023	Structural	Middle	Separates collaborative/visual/text views with contrastive learning
AlignRec [20]	2024	Mapping	Middle	Multi-stage alignment (within-content, content $\leftrightarrow$ ID, user $\leftrightarrow$ item) with pretrain + joint-train
DiffMM [18]	2024	Combined	Middle	Modality-aware graph diffusion with cross-modal contrastive learning
LGMRec [50]	2024	Structural	Middle	Local-global (hyper)graph learning disentangling collaborative vs. modality signals
MONET [69]	2024	Combined	Middle	Modality-embracing GCN with target-aware attention for dynamic weighting
PromptMM [19]	2024	Mapping	Middle	Prompt-tuning with multimodal knowledge distillation for efficient fusion
MLLM-MSR [21]	2025	Mapping	Middle	Uses multimodal LLMs for sequential recommendation
MENTOR [29]	2025	Mapping	Middle	Multi-level self-supervised objectives (feature & graph perspectives)
MoDiCF [76]	2025	Combined	Middle	Diffusion-based completion with counterfactual fairness constraints
SMORE [70]	2025	Structural	Middle	Spectral/frequency-domain alignment with adaptive fusion
LPIC [104]	2025	Mapping	Middle	Learnable prompts with ID-guided cross-modal alignment
AMMRM [41]	2025	Structural	Late	Modality noise filtering + adaptive user-preference fusion gate
MARGO [71]	2025	Mapping	Late	Modality-specific predictors with reliability-weighted late combination
MMAgentRec [30]	2025	Interactive	Late	LLM-orchestrated agentic loop integrating text, visual, and behavioral modules

modal information. MGAT [10] incorporates modality-aware attention into graph neighborhood aggregation, while GRCN [107] addresses interaction noise by refining the user-item graph structure before applying graph convolution, both optimized with pairwise ranking objectives.

While effective, these models tend to overfit in sparse or noisy feedback settings and often struggle to generalize when modality availability varies or shifts across domains [2, 5].

Table 5 Summary of Optimization Strategies in Multimodal Recommendation

Optimization Strategy	Contributions	Limitations
Supervised Learning [8] (2016), [10] (2020), [107] (2020), [11] (2021)	Leverages annotated interactions to train models with strong accuracy in data-rich environments.	Relies heavily on labeled data; performance drops in sparse or cold-start scenarios.
Self-supervised Learning [72] (2022), [64] (2025), [29] (2025)	Generates training signals from unlabeled data using contrastive or pretext tasks; robust to sparsity.	Effectiveness depends on the design of pretext tasks; may require careful modality balancing.
Distillation Learning [108] (2023), [19] (2024), [109] (2025)	Transfers knowledge from large pre-trained models to lightweight networks, enabling efficient inference.	Requires access to powerful teacher models and may incur performance degradation during distillation.

5.2 Self-supervised learning

Self-supervised learning (SSL) generates training signals from the data itself without requiring manual annotation. In MMRS, SSL objectives are typically constructed by creating augmented or complementary views of multimodal content, enabling models to learn robust cross-modal representations even when interaction supervision is sparse.

For instance, MMGCL [72] enforces consistency constraints across multimodal views through graph contrastive learning to enhance robustness in scenarios with sparse and missing modalities. DiffCL [64] employs a diffusion-based contrastive framework to generate augmented views and perform semantic alignment, thereby improving representation quality. MENTOR proposes a multi-level self-supervised framework that refines representations from feature and graph perspectives, enabling robust alignment across heterogeneous modalities [29].

SSL-based methods are particularly well-suited to pre-training multimodal encoders, offering reusable representations and greater adaptability to downstream personalization tasks [29, 78].

5.3 Distillation learning

Distillation learning facilitates efficient knowledge transfer from large, expressive teacher models (e.g., LLMs or multimodal encoders) to smaller, task-specific student networks. This paradigm is increasingly adopted to reduce model size and inference cost while preserving semantic richness, especially in cross-domain or low-resource recommendation scenarios.

PromptMM [19] employs prompt tuning and knowledge distillation to inject multimodal signals from frozen pre-trained encoders, improving fusion efficiency without full fine-tuning. Zhang et al. [109] address the domain gaps between pre-trained multimodal models and recommendation tasks, proposing adaptation strategies that complement distillation or prompt-based methods. Cross-Modal KD [108] uses optimal transport to align embedding dis-

tributions across modalities during knowledge distillation, thereby enhancing semantic consistency across heterogeneous inputs.

Despite these limitations, knowledge distillation is particularly valuable in MMRS deployment scenarios where large pretrained encoders provide strong multimodal representations but are too computationally expensive for real-time inference.

6 Evaluation

6.1 Datasets

Evaluation of MMRS relies heavily on publicly available real-world datasets across diverse application domains.

E-commerce and Product Recommendation. Among the most extensively studied are e-commerce datasets, where the Amazon Review datasets are widely used due to their large scale and rich multimodal content (user reviews, product images), and serve as standard testbeds in many recent works [35, 79, 98, 99, 110, 111].

Short-form Video Recommendation. Datasets such as MicroLens [112] and other large-scale short-video corpora [113] provide interaction logs, thumbnails, and textual metadata to support multimodal learning; TikTok-style micro-video platforms are also frequently used as benchmarks in practice [72, 107].

Movie and Entertainment Recommendation. MovieLens is often augmented with OMDb metadata (e.g., plot summaries, genres, poster images) to introduce multimodal inputs, enriching the original interaction matrix with visual and textual features [72].

News and Other Domains. News recommendation tasks are typically benchmarked on public datasets such as MIND and Adressa, which provide user click histories alongside titles, abstracts, and multimedia content (for news-model examples, see [114]). Additional domains include recipe-style multimodal setups (e.g., ingredients, instruc-

tions, images; see generative completion frameworks [76]) and multimodal user-interaction benchmarks [115].

Benchmarking Frameworks. To support standardized benchmarking, frameworks like Ducho-meets-Elliot provide unified evaluation pipelines, curated datasets, and consistent experimental settings tailored for MMRS research [116]. Most datasets include at least two raw modalities (text, images, or video) alongside user-item interactions (clicks, purchases, ratings), and are typically evaluated using Recall@K and NDCG@K, facilitating comparability across models and tasks.

6.2 Metrics

Evaluation of MMRS involves both conventional recommendation metrics and modality-aware diagnostic measures, reflecting the dual goals of prediction performance and effective multimodal fusion.

Traditional evaluation metrics primarily focus on the accuracy of recommendations and the quality of rankings. Precision and Recall assess the proportion of relevant items retrieved, while Hit Rate checks whether ground-truth items appear within the top-K recommendations. Mean Average Precision (MAP) and Normalized Discounted Cumulative Gain (NDCG) evaluate ranking quality, with NDCG placing greater emphasis on the rank positions of relevant items [117, 118]. In addition, beyond-accuracy metrics such as Diversity, Novelty, Coverage, and Popularity Bias are employed to evaluate recommendation fairness, personalization depth, and content diversity [119–121].

However, in multimodal settings, it is equally important to evaluate how modalities contribute to model performance. To this end, modality-level diagnostic metrics are increasingly adopted. Centered Kernel Alignment (CKA) [122] quantifies the similarity between modality-specific representations, whereas mutual information measures the amount of information shared across modalities. Orthogonality scores [123] help assess the independence or complementarity of modality embeddings. These metrics provide insight into whether the fusion mechanism is effectively integrating or redundantly overlapping multimodal inputs.

Furthermore, explainability and attribution methods play an important role in diagnosing MMRS behavior. Common approaches include ablation studies to assess the impact of each modality, Shapley value analysis to quantify contributions, and gradient-based attribution or attention-weight visualization to interpret decision-making. These tools are critical for identifying modality dominance, interaction bottlenecks, or underutilized information channels.

Effective MMRS evaluation demands a multi-faceted approach that combines conventional accuracy metrics with fusion-aware diagnostics. This dual focus enables researchers to assess not only what the system recommends but also how

different modalities interact and influence the final outcomes. As a quick reference, Table 6 summarizes commonly used accuracy, beyond-accuracy, and modality-diagnostic metrics with brief descriptions and representative references.

7 Challenges and future directions

Before discussing open challenges, we note a scope limitation: although the MMRS literature increasingly considers audio and video modalities, the vast majority of surveyed methods focus on text and image signals. Audio- and video-centric recommendation (e.g., music, podcast, short-video platforms) remains comparatively underexplored in the multimodal fusion literature, and we encourage future work to address this imbalance.

Cross-domain and Cross-platform Recommendation

MMRS face significant challenges in generalizing across domains and platforms due to differences in data modalities, user behavior patterns, and content distributions [106, 126]. These discrepancies lead to domain shifts and modality heterogeneity, which impair the transferability of learned representations. Most existing MMRS models are optimized within a single domain and struggle to adapt to unseen contexts without substantial retraining [126]. For instance, MuSiC [106] addresses cross-domain cold-start by learning target-domain feature distributions via diffusion, yet it still assumes partial modality overlap. To address this more broadly, future work should explore domain-invariant multimodal representation learning, modality-aware domain adaptation, and contrastive frameworks that account for domain semantics [108, 126].

Personalized modality preference modeling

Many approaches employ global or context-agnostic modality weighting [5]; however, fully user-specific and context-specific modality preference modeling remains underexplored. In practice, users exhibit varying preferences for modality. Specifically, some rely on visual signals, while others prioritize textual, audio, or contextual information [59, 94]. Ignoring these differences limits personalization and model interpretability. Future research should develop mechanisms for user-specific modality weighting, leveraging personalized attention networks [56], meta-learning-based adaptivity [91], and behavioral signal mining to dynamically adjust modality preferences. Such mechanisms would allow systems to tailor recommendations based on individual modality reliance, task context, and evolving user interests.

Table 6 Evaluation Metrics in Multimodal Recommendation Systems

Category	Metric	Description & Best For	References
Accuracy	Precision	Proportion of recommended items that are relevant; best for top-K exactness.	[117]
	Recall	Proportion of relevant items successfully retrieved; best for retrieval completeness.	[117]
	MAP	Mean of average precisions across users; best for overall ranking accuracy comparison.	[117]
	NDCG	Position-sensitive ranking quality; best when rank position matters (search, feed ranking).	[118]
	Hit Rate	Whether a relevant item appears in top-K; best for binary relevance in implicit feedback.	[117]
Beyond- Accuracy	Diversity	Dissimilarity among recommended items; best for evaluating breadth and exploration.	[119, 121]
	Novelty	Unexpectedness of recommendations; best for serendipity and long-tail coverage.	[119]
	Coverage	Fraction of the catalog ever recommended; best for detecting system-level popularity bias.	[120]
	Popularity Bias	Tendency toward popular items; best for auditing fairness and exposure equity.	[120, 121]
Modality Diagnostic	CKA	Similarity between modality-specific representation kernels; best for diagnosing fusion redundancy vs. complementarity.	[122]
	Mutual Information	Shared information between modalities; best for quantifying cross-modal overlap.	[4]
	Orthogonality Score	Independence of modality subspaces; best for verifying disentanglement quality.	[123]
	Shapley Value	Game-theoretic modality contribution attribution; best for fair, model-agnostic importance analysis.	[124]
	Gradient Attribution	Gradient-based feature impact estimation; best for instance-level interpretability.	[125]
	Attention Weights	Learned attention scores for modality importance; best for lightweight interpretability.	[56]

Multimodal knowledge graphs and commonsense reasoning

Knowledge graphs have enhanced explainability and improved structure awareness in traditional recommendation systems [6, 53]. However, their use in multimodal settings remains limited. A multimodal knowledge graph (MMKG) integrates both symbolic semantics and perceptual evidence, linking entities to images, audio, and video features [127]. Although MMKGs already exist, a central challenge remains cross-modal alignment: consistently linking KG entities to the appropriate perceptual signals when data is missing, updates are out of sync, or the domain changes [6]. Most current MMRS models also lack commonsense reasoning, which is important for understanding content and user intent. Future works can construct MMRS-oriented MMKGs that encode semantic and perceptual relationships with temporal consistency and reliable cross-modal alignment, and draw on

external resources such as Visual Genome, Conceptual Captions, and ATOMIC to inject commonsense priors [6, 127].

Integrating LLMs into MMRS pipelines

LLMs provide powerful tools for natural language understanding, semantic alignment, and cross-modal reasoning; however, their application in recommender systems remains relatively nascent [22]. Current LLM-based MMRS applications often treat LLMs as feature extractors or explanation generators [19, 63], but they are increasingly being integrated for generative retrieval and agentic decision-making [30, 102, 103]. Large-scale ranking integration is still emerging, and ensuring factual grounding, user trust, and domain relevance of LLM-generated rationales remains an open challenge [22]. Future research should focus on adapting LLMs to the recommendation objective, designing lightweight fusion and alignment modules, and developing scalable,

recommendation-aware techniques for modality-conditioned generation and reasoning [21, 29].

Cross-domain fairness in multimodal recommendation

Cross-domain MMRS encounters serious fairness issues [76, 126]. When transferring knowledge across domains, existing differences in user communities, engagement patterns, and item popularity tend to worsen, creating unequal exposure and biased results. For example, MoDiCF [76] combines diffusion-based modality completion with counterfactual re-ranking to mitigate visibility bias caused by missing modalities, but broader cross-domain fairness remains under-addressed. Future work should investigate adding fairness constraints for users and items during the transfer process, as well as developing adjustment techniques that can handle changing data patterns [76, 120]. These approaches would enable cross-domain multimodal recommendation systems to function effectively across different domains while maintaining fairness and responsibility.

8 Conclusion

This survey synthesizes over 127 publications spanning 2016–2025 and introduces a pipeline-based taxonomy that decomposes MMRS into six interconnected stages: modality extraction, filtration, bridging, fusion, augmentation, and optimization. Unlike prior surveys that focus predominantly on model architectures [5] or application domains [1], our framework emphasizes the intermediate processing stages, i.e., filtration (noise removal, missing-modality handling, efficiency screening) [15, 45, 47], bridging (structural, interactive, mapping, combined) [9, 10, 28], and augmentation (data-, feature-, representation-level) [29, 78]. These are critical for robustness yet under-analyzed in the literature.

For each stage, we systematically review representative methods, summarize design motivations, and illustrate use cases. We extend the evaluation discussion beyond conventional accuracy metrics by introducing modality-specific diagnostics, such as CKA [122], Shapley-value attribution [124], and gradient-based interpretability [125], as recommended tools for assessing how effectively multimodal signals are integrated within recommendation pipelines.

Looking forward, we identify several promising research directions, including cross-domain generalization [126], adaptive modeling of user-specific modality preferences [94], integration with multimodal knowledge graphs and common-sense reasoning [6, 127], the scalable adoption of LLMs and foundation models in MMRS workflows [21, 22], and cross-domain fairness in MMRS [76, 126]. Addressing these challenges will be essential to advancing MMRS toward

systems that are more accurate, robust, interpretable, and equitable across diverse users and content domains.

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